



## 0. Loops (scenario-driven, not pattern-printing)

- **LC 1360 — Number of Days Between Two Dates** — calendar math forces careful loop/edge-case thinking (leap years). Very "real world."
- **LC 1518 — Water Bottles** — bottle exchange simulation. Simple loop, but the "when do I stop" condition trips people up in a good way.
- **LC 1103 — Distribute Candies to People** — zig-zag distribution scenario, makes them simulate rather than formula-jump.

## 1. Arrays

- **LC 1652 — Defuse the Bomb** — circular array indexing with a fun "bomb defusal" framing; good discussion on modulo wraparound.
- **LC 1275 — Find Winner on a Tic-Tac-Toe Game** — engaging simulation, forces structured condition-checking instead of brute state comparison.
- **LC 665 — Non-decreasing Array** — deceptively easy-sounding; the edge cases around "which element to fix" spark real debate.

## 2. String

- **LC 819 — Most Common Word** — banned-word text processing; nice blend of string cleanup + hashing, feels like a real mini-project.
- **LC 1189 — Maximum Number of Balloons** — frequency-counting with a twist (multiple letters needed per occurrence) — good "wait, that's not just a count" moment.
- **LC 1935 — Maximum Number of Words You Can Type** — broken-keyboard scenario, quirky and memorable.

## 3. Hashing

- **LC 205 — Isomorphic Strings** — bidirectional mapping requirement is the real teaching point (one-way map isn't enough) — strong "aha."
- **LC 290 — Word Pattern** — same core idea as Isomorphic Strings but with words instead of characters; good for reinforcing the concept from a different angle.
- **LC 1002 — Find Common Characters** — "shopping list intersection" framing, frequency-array-per-word thinking.

## 4. Binary Search (beyond "search a sorted array")

- **LC 441 — Arranging Coins** — staircase scenario, binary search *on the answer* rather than on an array — this reframing is usually the biggest lightbulb moment of the day.
- **LC 540 — Single Element in a Sorted Array** — clever index-parity insight; genuinely surprising that binary search applies here at all.
- **LC 367 — Valid Perfect Square** — another "binary search on the answer" case, easy to grasp, reinforces the pattern from 441.



## 5. Sort

- **LC 922 — Sort Array By Parity II** — quick warm-up, index-placement thinking rather than a generic sort call.
- **LC 179 — Largest Number** — build the largest number from digits using a custom comparator. This is the standout problem of the set — genuinely non-obvious why the comparator works, great whiteboard discussion.
- **LC 1029 — Two City Scheduling** — cost-based greedy scenario (fly half to city A, half to city B); real decision-making framing, not just "sort ascending."